



Ecozen Solutions is the brainchild of three IIT-Kharagpur geniuses, who are using solar-powered solutions to increase irrigation and reduce food wastage across India's farmlands

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According to the Indian government's official figures from June 2021, agriculture and allied sectors still contribute over 20 per cent to India's GDP. Needless to say, modernising Indian agriculture through technology will not only benefit the farmers practicing the trade but also the nation, at large, in terms of ensuring sustainable, long-term food security. That's exactly what three IIT-Kharagpur graduates had in mind, when they decided to start Ecozen Solutions.

"Increasing access to climate-smart technology is what drives us at Ecozen," Co-Founder & CEO, Devendra Gupta told us, underscoring how their agri-tech products have demonstrated the impact clean energy can have on the agricultural sector in the past 12 years.

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Needless to say, agriculture in India needs all the help it can get, because it's plagued by a few fundamental

problems – lack of access to consistent electricity supply and high operational costs. These issues were primarily the reason for only 34.3 per cent of arable land being irrigated in India, which ultimately resulted in lower farm productivity and lower incomes for farmers, according to Gupta. On top of that only 27 per cent of farmers in UP were aware of solar pumps and only



Devendra Gupta, Co-Founder & CEO, Ecozen

10 per cent believed they were a viable solution, as per a 2018 CEEW report.

Issues related to growing crops in agricultural land notwithstanding, Ecozen's founders were also aware of another huge problem of farm waste. Believe it or not, but nearly 40 per cent of agricultural produce gets wasted in India – even before it gets to market, claimed Gupta. This is primarily due to a lack of access to a cold chain (only 4 per cent of the produce that needs a cold chain has access to it) and proper post-harvest management (PHM) operations. These are some of the core problems Ecozen Solutions is trying to solve. With a 20 per cent market share, Ecozen is leading the adoption of sustainable technology in India's agricultural sector.

"With our technology focus, R&D is a core element of our process at Ecozen," underscored Devendra Gupta. "This has seen us bring a few technology firsts to the solar pumping segment, with innovations in connectivity, advanced motor controls, AI and IoT. The most recent of these innovations is the 4G network connectivity of our